◈ **RetailIQ**

RETAIL INTELLIGENCE PLATFORM

NAVIGATION

- Executive Overview
- Demand Forecasting
- Historical vs Forecast
- Model Performance
- Store & Category Analysis
- Inventory Optimization
- High-Risk Products

FILTER DATA

Product Category

HOUSEH... × FOODS × HOBBIES ×

Store

CA_1 × CA_2 × CA_3 × WI_1 × WI_2 × WI_3 ×

Search Product

Search product ID...

RETAIL DEMAND & INVENTORY INTELLIGENCE

Smarter Demand. Better Inventory Decisions.

A data-driven view of retail demand, model performance, product trends and inventory risk.

Displaying 30,490 product-store combinations across 10 stores.

Model Performance

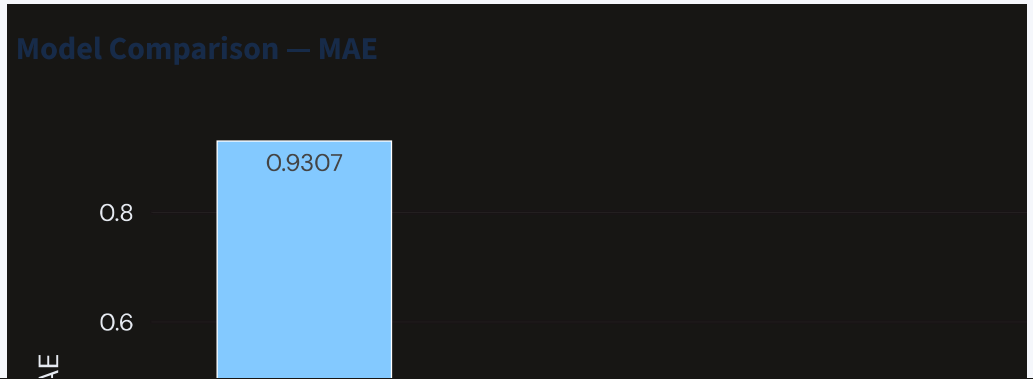
Compare the validation performance of the machine learning models developed for retail demand prediction.

Best MAE	Best RMSE	Best WAPE
0.8879	2.0061	68.37%
↑ LSTM — Epoch 5	↑ LightGBM	↑ LSTM — Epoch 5

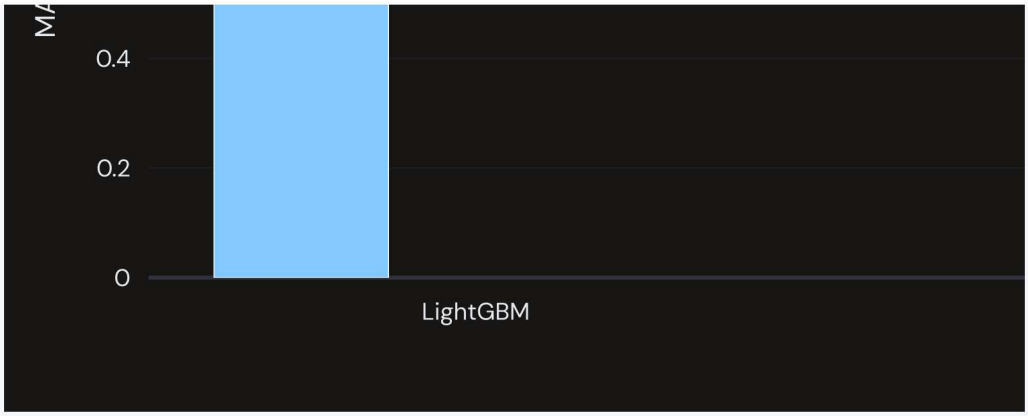
Validation Metric Comparison

Select Metric

MAE



TX_1 ×
TX_2 ×
TX_3 ×



Model Scorecard

Model	MAE
LightGBM	
LSTM — Epoch 2	
LSTM — Epoch 5	

How to interpret these results

MAE measures average absolute prediction error. RMSE penalizes larger errors more heavily. WAPE expresses aggregate absolute error relative to actual demand. Lower values indicate better performance for each metric.

A model may perform better on one metric and worse on another, so model selection should consider the business objective.

RetailIQ
Retail Demand Forecasting & Inventory Intelligence

Built with Python · Pandas · LightGBM · PyTorch · Streamlit · Plotly
Historical M5 benchmark | May 2016 – June 2016